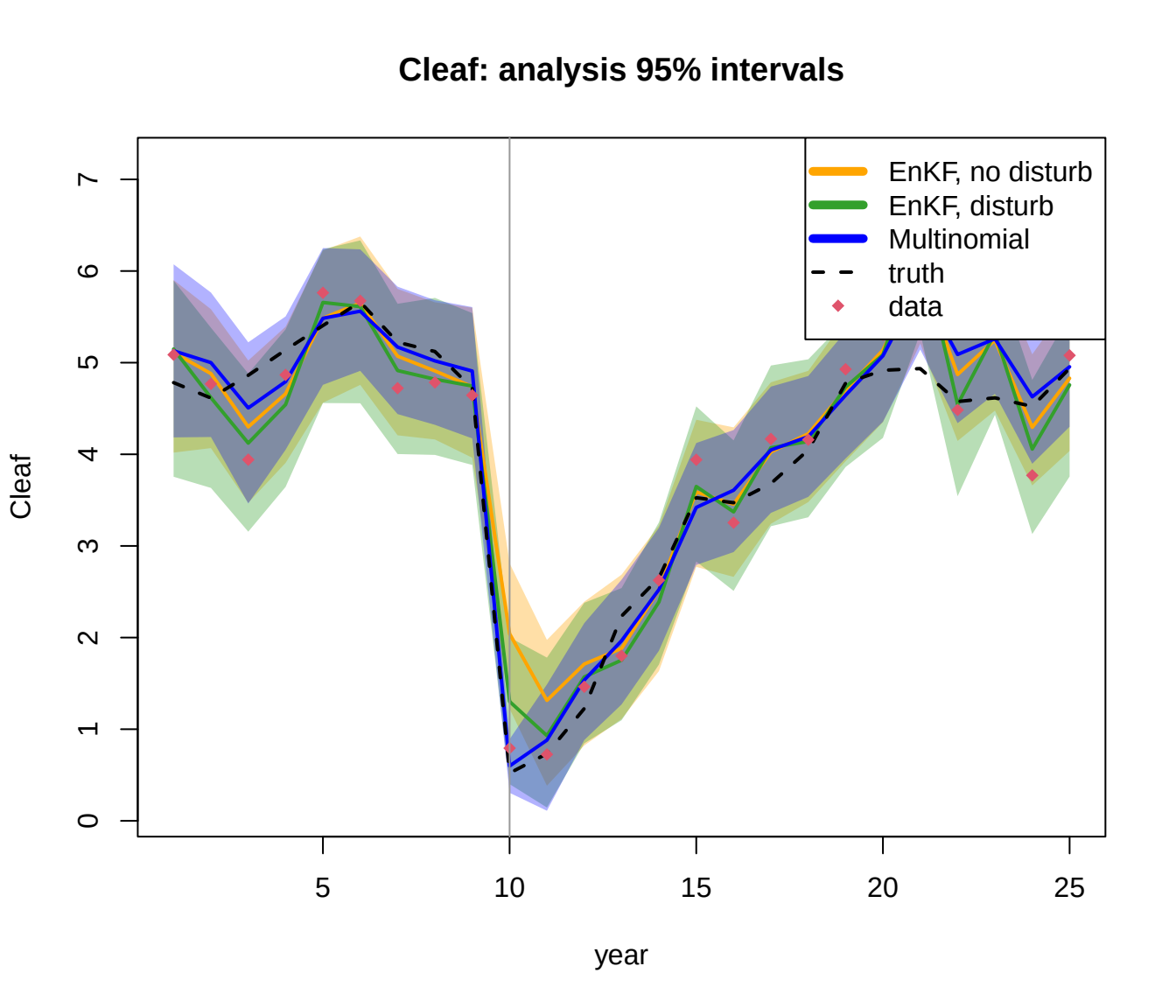
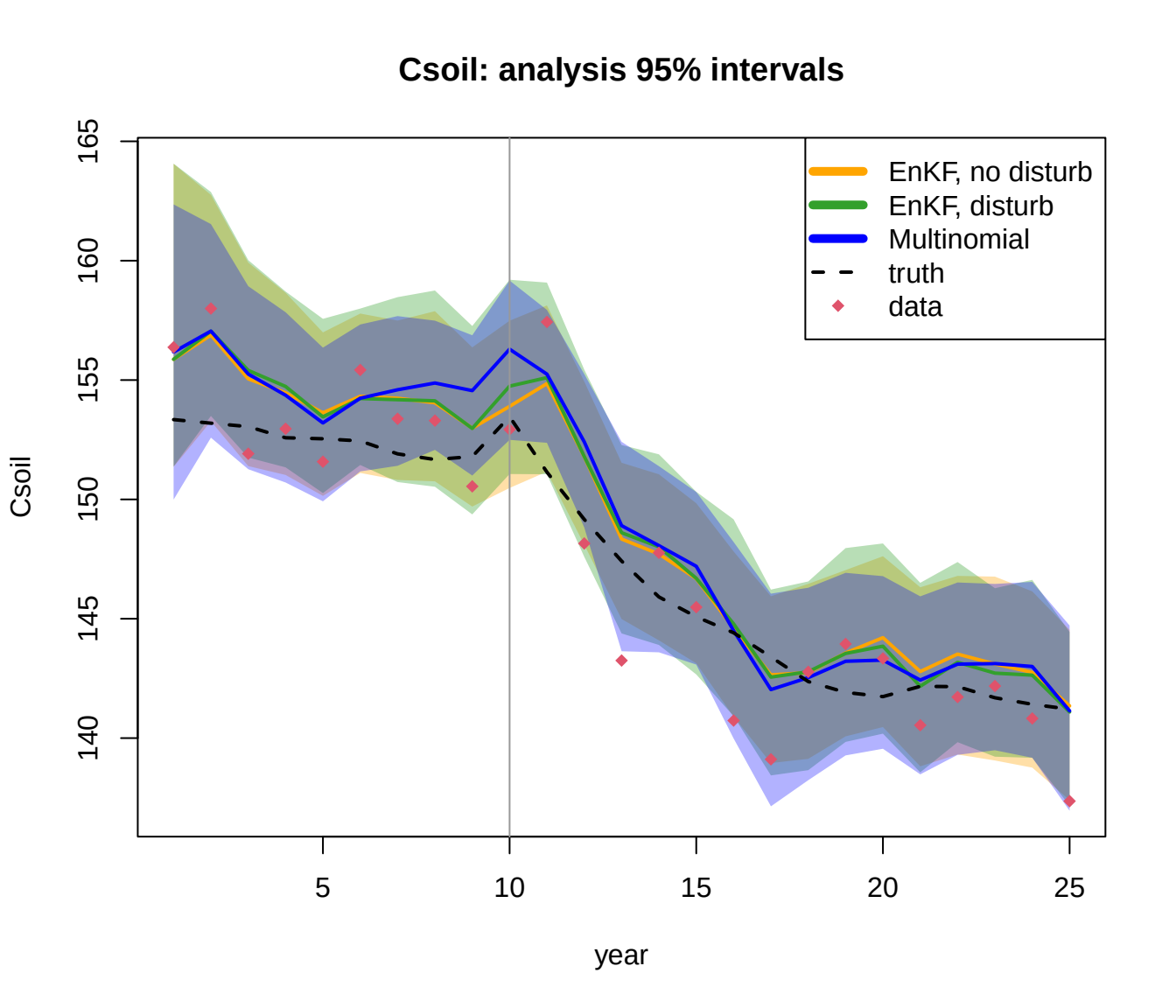
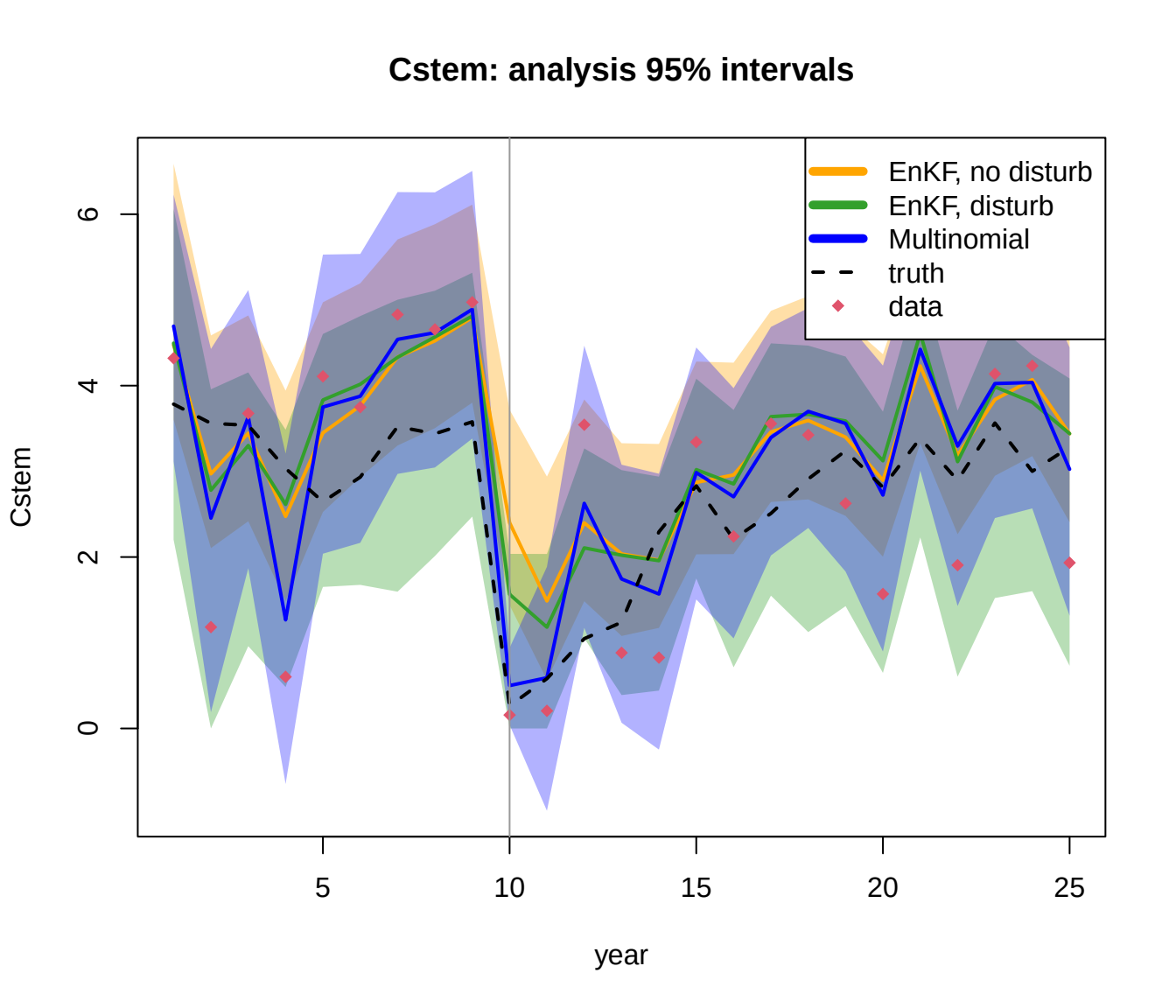
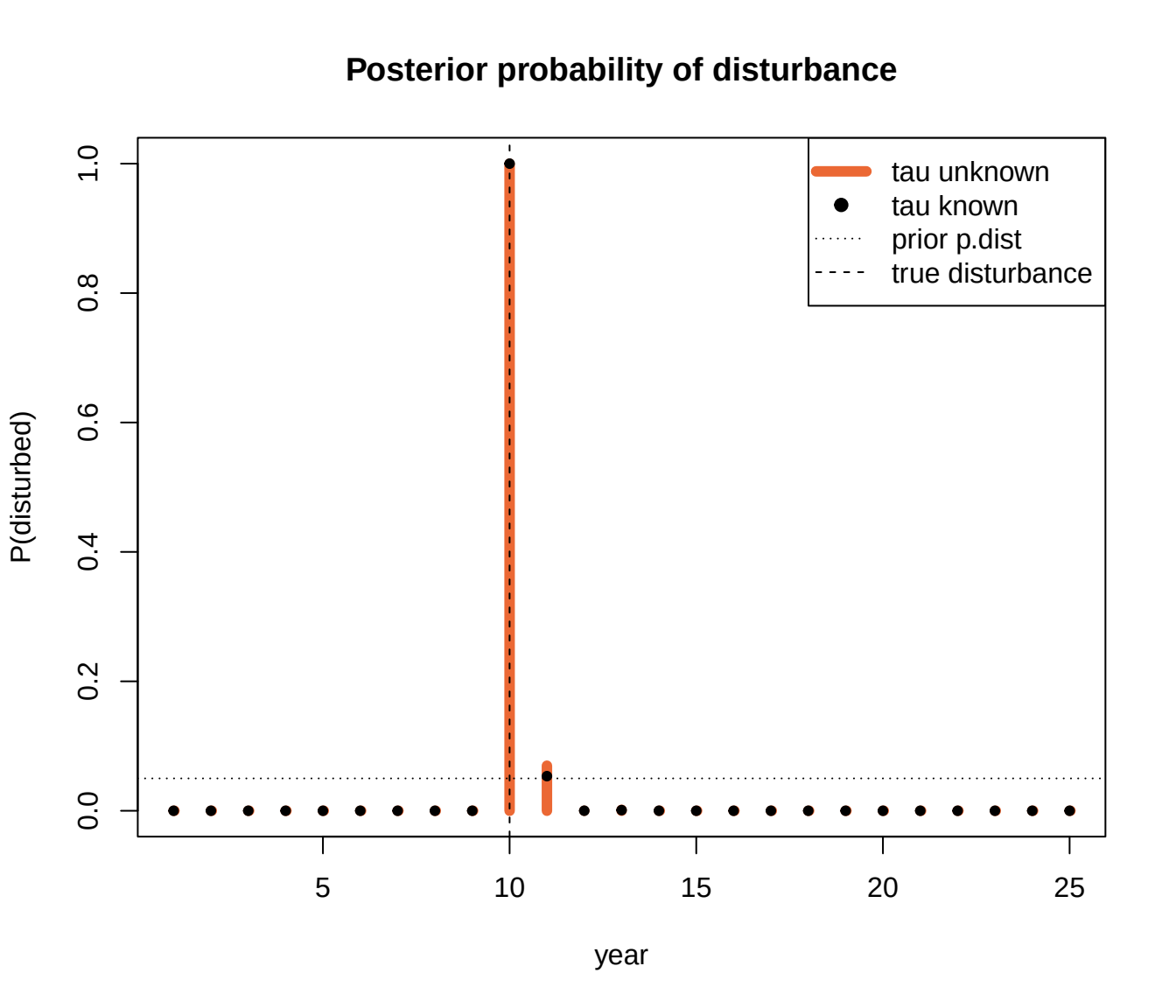
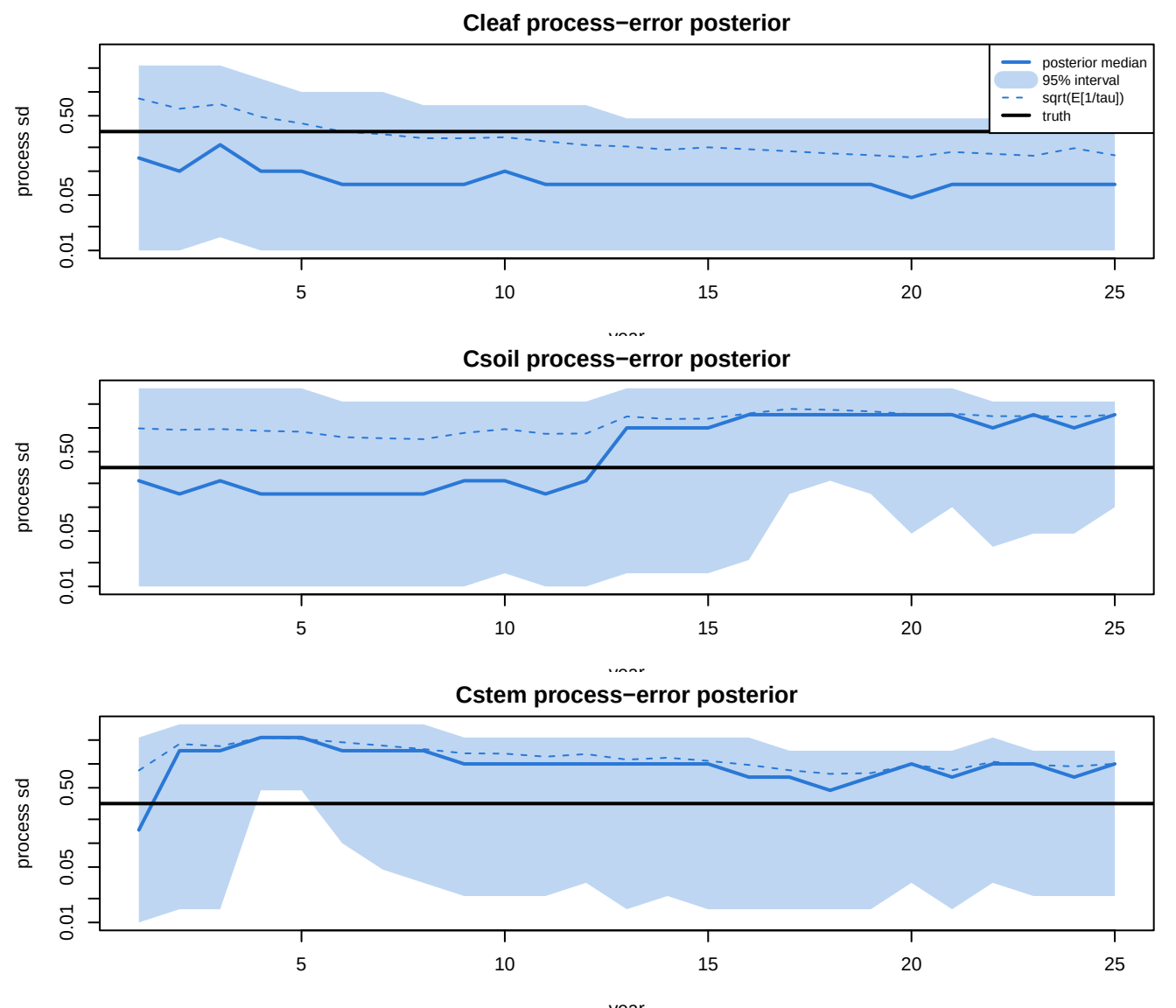
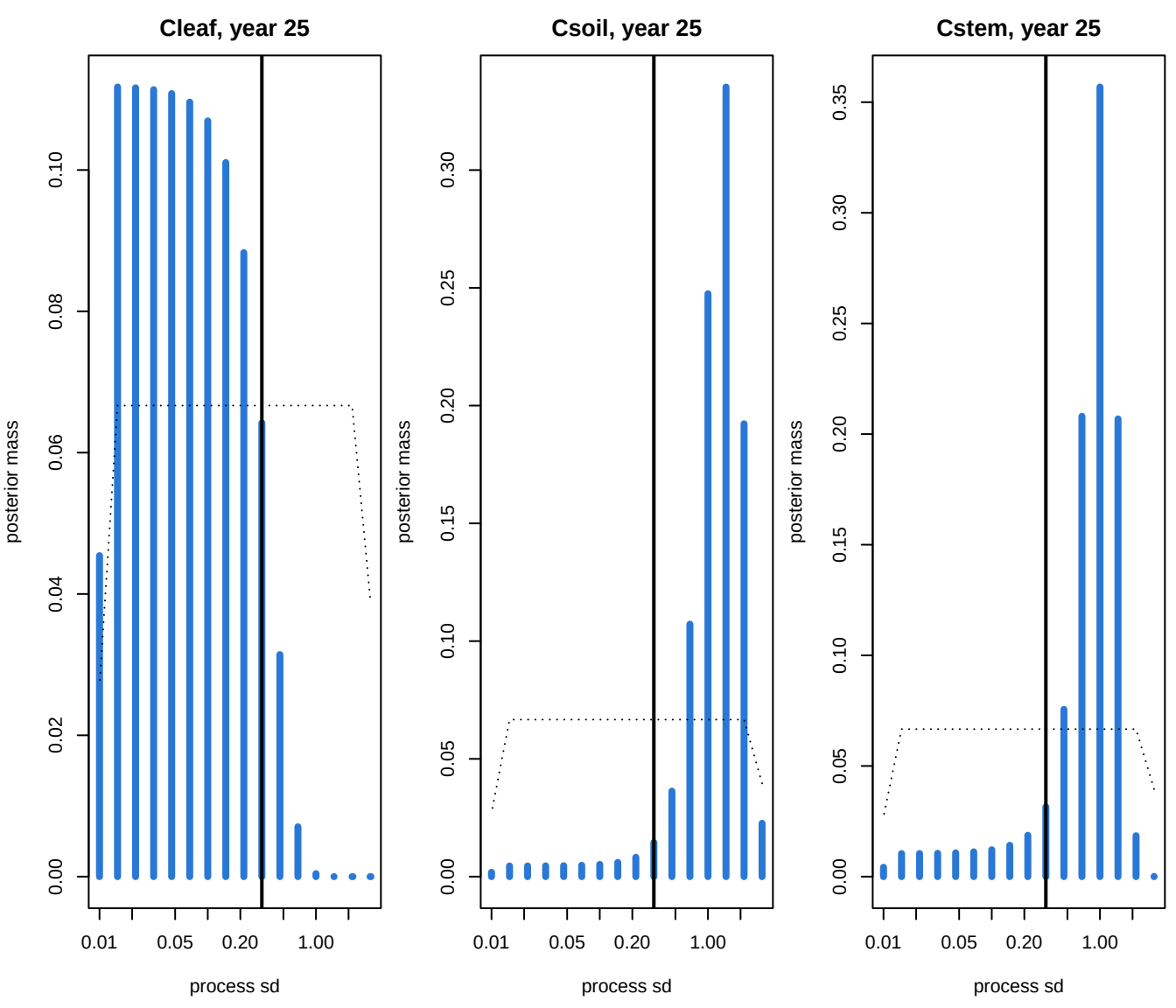

Cleaf: analysis 95% intervals
EnKF, no disturb
EnKF, disturb
Multinomial
truth
data
Cleaf
0
1
2
3
4
5
6
7
5
10
15
20
25
year

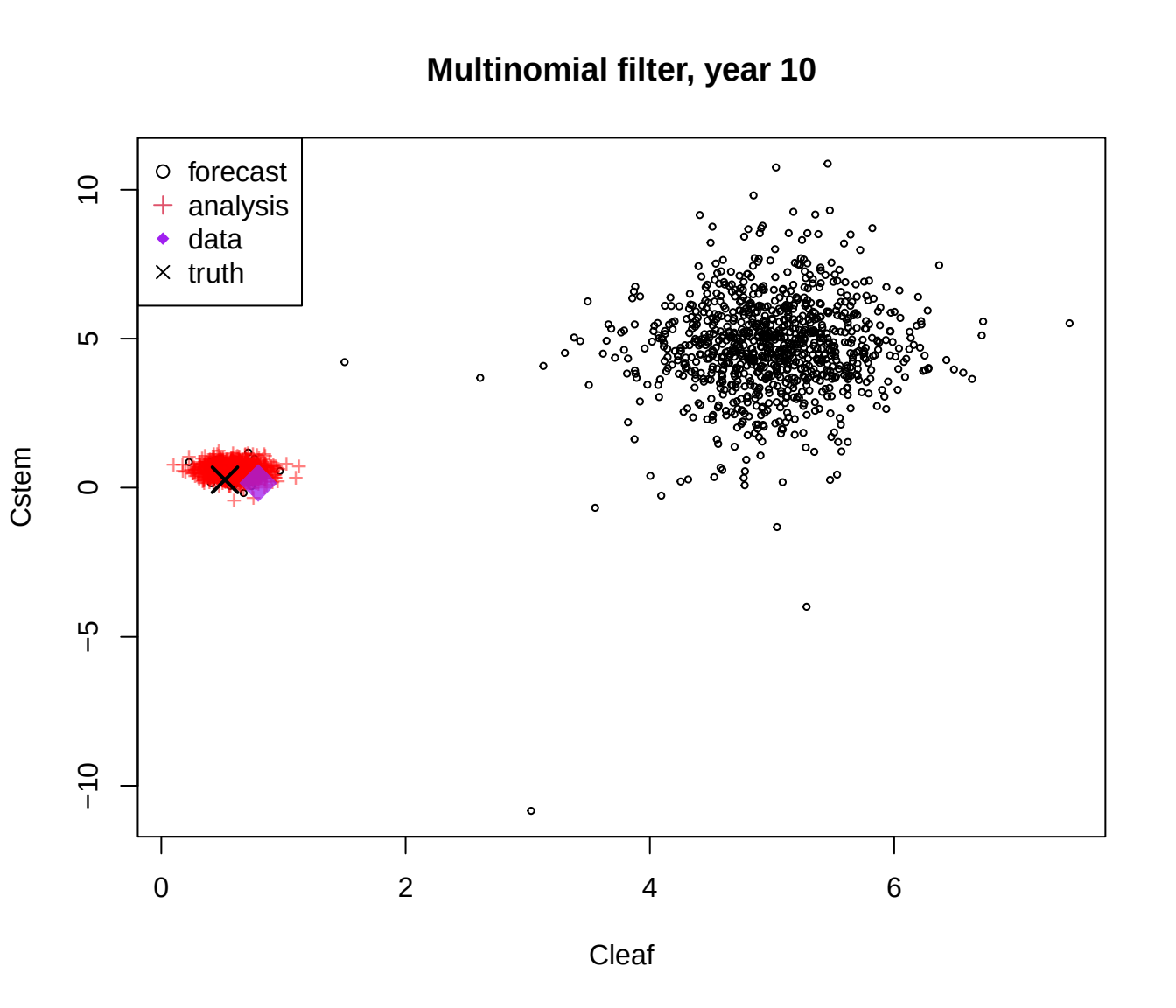

Csoil: analysis 95% intervals
EnKF, no disturb
EnKF, disturb
Multinomial
truth
data
Csoil
year
140
145
150
155
160
165
5
10
15
20
25


Cstem: analysis 95% intervals
EnKF, no disturb
EnKF, disturb
Multinomial
truth
data
Cstem
0
2
4
6
5
10
15
20
25
year


Posterior probability of disturbance
tau unknown
tau known
prior p.dist
true disturbance
P(disturbed)
0.0
0.2
0.4
0.6
0.8
1.0
5
10
15
20
25
year


Cleaf process−error posterior
process sd
0.01
0.05
0.50
5
10
15
20
25
posterior median
95% interval
sqrt(E[1/tau])
truth
Csoil process−error posterior
process sd
0.01
0.05
0.50
5
10
15
20
25
Cstem process−error posterior
process sd
0.01
0.05
0.50
5
10
15
20
25


Cleaf, year 25
Csoil, year 25
Cstem, year 25
posterior mass
process sd
0.01
0.05
0.20
1.00
0.00
0.02
0.04
0.06
0.08
0.10
0.05
0.10
0.15
0.20
0.25
0.30
0.35


Multinomial filter, year 10
forecast
analysis
data
truth
Cstem
10
5
0
−5
−10
0
2
4
6
Cleaf